

Elastic constants of - GeO 2

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Elastic constants of α -GeO₂

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The elastic constants of α -GeO₂, the quartzlike polymorph of germanium dioxide, have been determined using Brillouin scattering. The data were obtained using a high quality single crystal of α -GeO₂. When compared with existing data on α -SiO₂, our results show that the shear constants are considerably softened in GeO₂. This could be an indication that the observed pressure induced phase change is related to a shear instability. © 1998 American Institute of Physics.

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I. INTRODUCTION

To understand high pressure behavior of materials, it is often useful to consider the trends in Mendelev's periodic table: moving down the columns in Mendelev's table is, in many respects, similar to increasing pressure. In this respect, and given the fundamental importance of SiO₂ in geophysics, the comparison of the properties of GeO₂ and SiO₂ cannot be overestimated.

GeO₂ at normal pressures has two stable modifications. The low temperature form has the rutile structure (tetragonal with $a=4.395$ Å, $c=2.86$ Å, and $\rho=6.28$ g/cm³). At $T \approx 1300$ K, it smoothly converts to a β -quartzlike structure which can be easily quenched to the α -quartzlike phase (hexagonal with $a=4.29$, $c=5.65$, and $\rho=4.28$ g/cm³). The existence of two stable forms of GeO₂ at one atmosphere can be understood by noting that the ratio of ionic radii $R(\text{Ge}^{4+})/R(\text{O}^{2-})=0.43$, is very close to the critical ratio 0.44 at which the change from four to sixfold coordination occurs.¹ The average interatomic distance Ge–O (1.65 Å) in α -GeO₂ and the bond angle (106.3–113.1°) at the Ge atom, is evidence for the distortion of the ideal tetrahedral symmetry.

It was the existence of GeO₂ in its tetragonal form which led to the idea that such a phase might also exist for SiO₂ at high pressures. Such a transformation, quartz-stishovite, was subsequently discovered by Stishov and Popova.² Conversely, it is clear that the properties of α -GeO₂ at even moderate pressures can greatly enhance our understanding of α -SiO₂ at very high pressures.

In spite of being one of the most widely investigated materials, SiO₂ still possesses properties which are not fully understood. Recently, phenomena related to its behavior under the very high pressures, now routinely achievable with diamond anvil cells, have attracted particular attention. Pressure induced amorphization has been found in both the quartz^{3–5} and coesite⁵ polymorphs of SiO₂. The mechanism

responsible for the crystal to amorphous transformation is, however, still under debate. Although considerably less investigated, GeO₂ has shown many of the same unusual properties exhibited by SiO₂; the amorphous form undergoes irreversible densification under pressure,^{6–8} the α -polymorph is also known to amorphize under pressure,^{9–12} the Ge changes from four to sixfold coordination at high pressures,¹³ and the sound velocity in amorphous GeO₂ shows irreversible changes with pressure.¹⁴

One of the controversial properties of quartz SiO₂ is the elastic anisotropy of the pressure amorphized material.¹⁵ It was reported in Ref. 15 that the sound velocities, in the pressure induced amorphous phase, along the original x and z axes of the crystal, are different. In Ref. 15, this difference was interpreted as due to anisotropy in the amorphous phase. This interpretation has since been debated in the literature^{16,17} and the issue is still unresolved: the authors of Ref. 15 believe it to be an intrinsic effect of the amorphous phase, those of Ref. 16 believe it is due to crystalline contamination. Although similar experiments have been performed on the sister compound AlPO₄,¹⁸ this material recrystallizes on pressure release making it even harder to ascertain the origin of the anisotropy. Before a similar investigation can be performed on GeO₂, it is necessary to understand the elastic properties of untransformed material; in this investigation, we determine the elastic constants of GeO₂ using Brillouin scattering techniques.

The unavailability of high quality single crystals of α -GeO₂ has precluded such investigations in the past. All reported work on α -GeO₂ to date has been on powder samples.^{9–12,19} High pressure studies on single crystals are currently in progress and will be published elsewhere. Here we present the results of a Brillouin scattering determination of the room temperature elastic constants at one atmosphere. The single crystal sample used in our study was fabricated by a novel hydrothermal technique described in detail in

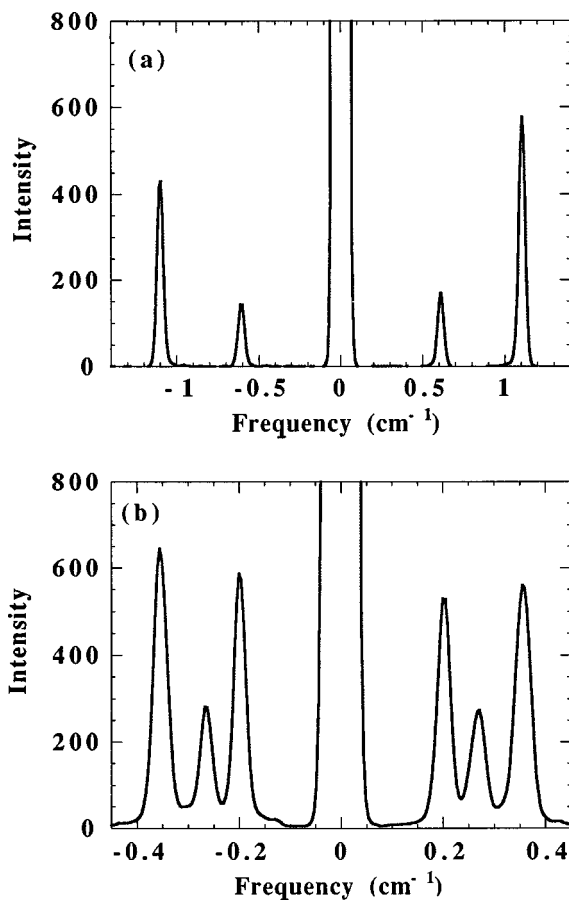


FIG. 1. Brillouin spectra of GeO_2 . (a) was recorded in the backscattering geometry with a 5+4 pass interferometer. (b) Corresponds to a platelet geometry and was obtained with a 5+2 pass instrument.

Refs. 20 and 21. The particular piece used in our experiments was a platelet with the c axis along the normal.

II. RESULTS AND DISCUSSION

Brillouin scattering is the inelastic scattering of light by acoustic phonons. A general description of the technique and instrumental details are given in the review article by Sandercock.²² For the purpose of this manuscript, the basic principles of Brillouin scattering and the scattering geometries used in this work can be found in Ref. 23. The scattering geometries available to us with this sample²³ were backscattering and platelet. The former allowed us to probe propagation directions close to the c axis while the latter enabled propagation directions in the basal plane to be investigated.

Figure 1 shows typical spectra obtained in the two scattering geometries with a 5+2 or 5+4 pass tandem Fabry-Perot interferometer.²² In backscattering [Fig. 1(a)], we observe the longitudinal ($\approx c_{33}$) and (degenerate) transverse ($\approx c_{44}$) modes; in platelet [Fig. 1(b)], we observe the longitudinal ($\approx c_{11}$) and both transverse modes ($\approx c_{44}$ and $\approx c_{66}$). The remaining two constants, c_{13} and c_{14} , enter the expressions describing the propagation dependence out and in plane, respectively.

The equations relating the measured frequency shifts to velocities (V) and elastic constants^{22,23} contain the density

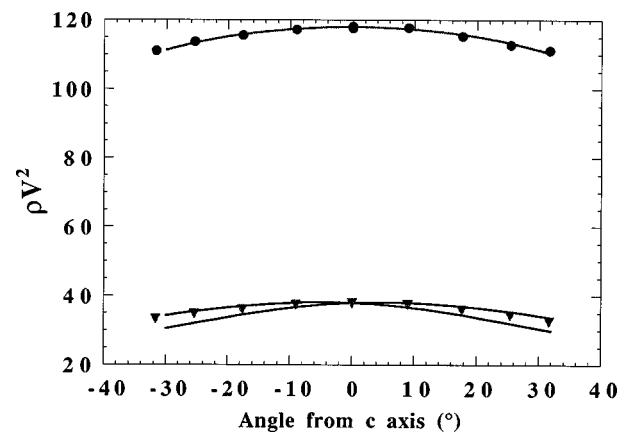


FIG. 2. ρV^2 obtained in the backscattering geometry as a function of the angle between the phonon propagation direction and the c axis.

(ρ) and refractive index. In our data analysis, we have used a refractive index ($n=1.65$) measured in our laboratories. The density we measured pycnometrically, ($\rho=4.3\pm 0.1 \text{ g/cm}^3$) is in good agreement with x-ray determinations ($\rho=4.28 \text{ g/cm}^3$). Figure 2 shows ρV^2 obtained from our backscattering experiments and plotted as a function of the phonon propagation angle away from the c axis. Along the c axis, the values of ρV^2 for the longitudinal (L) and transverse (T) are directly c_{33} and c_{44} . From the angle dependence in principle, it is possible to obtain c_{11} and c_{13} . It turns out however that the fits are not very sensitive when both are allowed to vary. We therefore fixed c_{11} at the value obtained from platelet data and adjusted c_{13} to fit the data in Fig. 2; since it was not possible to identify which of the two transverse branches correspond to the data, we treated the experimental results as an average of both branches. The constants extracted from Fig. 2 are listed in the first row of Table I. The full lines in Fig. 2 are the calculated ρV^2 using the c_{ij} from Table I. Figure 3 shows the dependence of ρV^2 on in-plane angle. The in-plane velocities do not depend on c_{33} and c_{13} , and for crystals of this symmetry it must be recalled that $c_{66}=(c_{11}-c_{12})/2$. Furthermore the lack of any clear variation with angle indicates that the constant c_{14} is almost zero. A least squares fit to the data yields the values listed in the second row of Table I. Note that if c_{14} were zero, the values of ρV^2 for the three modes would be given exactly by c_{11} , c_{44} , and c_{66} . The full lines in Fig. 3 are calculated using the c_{ij} from Table I.

Also listed in Table I are the c_{ij} of $\alpha\text{-SiO}_2$. Since the bulk modulus (B) of a trigonal crystal is given by

TABLE I. Elastic constants (in GPa) of $\alpha\text{-GeO}_2$ obtained from backscattering and platelet Brillouin experiments. c_{ij} of SiO_2 are included for comparison.

	c_{11}	c_{33}	c_{12}	c_{13}	c_{44}	c_{66}	$ c_{14} $	B
Back		118		32	38			
Plat	64		22		36	21	2	
Comb.	64	118	22	32	37	21	2	42
SiO_2	87	107	7	13	57	40	18	38

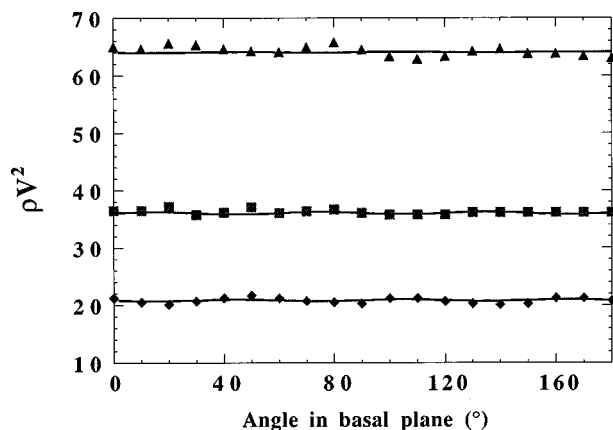


FIG. 3. ρV^2 obtained in the platelet geometry as a function of angle in the basal plane.

$$B = \{(c_{33}(c_{11} + c_{12}) - 2c_{13}^2)/(c_{11} + c_{12} + 2c_{33} - 4c_{13}), \quad (1)$$

we also include the values of B in Table I. The most notable differences between GeO_2 and SiO_2 are in c_{12} and c_{13} which differ by a factor of 3 and 2.5, respectively, and are larger in GeO_2 . This contrasts with the 35%–50% decrease in the shear constants c_{44} and c_{66} . Last, the small value of c_{14} in GeO_2 makes it almost elastically “hexagonal”.

Our value of B for GeO_2 (42 GPa) can be compared with those obtained from the pressure dependence of the lattice constants^{24–26} which yield 32.8, 39.4, and 63.9, respectively. The two latest determinations^{24,25} are consistent with our result; agreement can even be improved if the derivative of B with pressure (B') is reduced to ≈ 3.5 from the value 6 and 3.8 obtained in Refs. 24 and 25.

III. CONCLUSIONS

The trends between the c_{ij} of GeO_2 and SiO_2 can be used to evaluate the conditions for mechanical stability viz.: $c_{11} - c_{12} > 0$, $(c_{11} - c_{12})c_{44} - 2c_{14}^2 > 0$, and $(c_{11} + c_{12})c_{33} - 2c_{13}^2 > 0$. The first two correspond to shear instabilities, the third expression corresponds to a volume collapse. While the last condition is not very different for the two materials, the first two are almost a factor of 2 smaller in GeO_2 . Although far from conclusive, this may be an indication that the pressure induced phase transition in these materials is due to a shear instability.

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